

Lecture 14

Markov Perfection

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Basic Framework

- Society composed of two groups: Rich and Poor
- Rich are in power, Poor are majority
- If democratize, Poor are in power
- Whoever is in power redistributes in each period
- When Rich are in power, Poor can mount a revolution

Mechanism for Democratization

- Some periods are more favorable for revolution than others
- In favorable periods, Rich want to redistribute a lot
- May not be able to do enough in one period (credit constrained)
- Cannot credibly commit to future redistribution because only want to redistribute if period is favorable to a revolution
- Democratization is a means of credible commitment to redistribution

Players

- Two types of agents: Poor and Rich
- A share $\lambda \in (\frac{1}{2}, 1)$ are poor
- Continuum of agents of mass 1
- Each period, an agent from group G has capital h^G
 - $h^R > h^P \geq 1$
 - Total capital is $H = \lambda h^P + (1 - \lambda)h^R$

Production and Taxation

- Two production technologies
 - Market production: $Y_t^m = AH_t^m$
 - Home (Non-Market) production: $Y_t^n = BH_t^n$
- Market production more productive: $A > B$
- The government can impose a linear tax only on market production
 - For tax rate τ_t , government revenue is $\tau_t Y_t^m$
 - Equal transfer, $T_t = \tau_t Y_t^m$, to each person
 - Government cannot target taxes or transfers
 - Government budget constraint is $T_t = \tau_t AH_t^m$

Payoffs from Production

- All agents have linear preference over income and discount future by $\delta \in (0, 1)$.
- Post tax income to player i in period t is

$$\hat{y}_t^i = (1 - \tau_t)Ah_i^m + Bh_i^n + T_t$$

Revolution

- Poor agents can overthrow the government ($r = 1$) with certainty if they choose to
- If revolt, poor take control of all capital
- A fraction $1 - \mu_t$ is destroyed
- If a revolution occurs, each poor agent receives a per-period return of $\frac{\mu_t AH}{\lambda}$
- μ is stochastic

$$\mu = \begin{cases} \mu^h > 0 & \text{with probability } q \\ \mu^l = 0 & \text{with probability } 1 - q \end{cases}$$

Interpretation

- Amount destroyed may be stochastic for a variety of reasons
 - Government or insurgent capacity
 - Weather
 - International factors
- More generally, the opportunity costs of revolution may be stochastic for many reasons

Timeline of Stage Game

1. State μ_t is revealed
2. Elite decide whether to democratize ($F_t = D$). If not $F_t = E$, set tax rate.
3. Poor decide whether to revolt, If not, and $F_t = D$, they set tax rate.
4. Agents allocate capital between modes of production and incomes realized.

Equilibrium Concept

- Restrict attention to Markov Perfect Equilibrium
- Agents only condition behavior on *payoff relevant* factors
- Here, that is μ and whether or not there has been democratization
- In any subgame in which those two facts are the same, a player plays the same. No history-dependent punishment strategies.
- This is like stationarity from Baron-Ferejohn, only with some payoff relevant facts from the history

Taxation and Production

- The payoff from allocating a unit of resources to the market is $(1 - \tau)A$
- The payoff from allocating a unit of resources to home production is B
- Allocate all to the market in t if $(1 - \tau_t)A \geq B$ or

$$\tau_t \leq \frac{A - B}{A} \equiv \hat{\tau}$$

- No voter (Rich or Poor) would ever choose $\tau_t > \hat{\tau}_t$
- Restrict attention to $\tau_t \leq \hat{\tau}$ and all resources in market production

Unitary Actors

- All Poor agents have identical preferences
- All Rich agents have identical preferences
- Treat these groups as unitary actors, without loss of generality
 - This abstracts away from some of the interesting microfoundations of revolution

Attempting Revolution

- Poor will only consider attempting revolution if $\mu_t = \mu^h$
- If revolution, continuation value for poor is

$$W^P = \frac{\mu^h AH}{\lambda(1 - \delta)}$$

- Rich lose everything, so continuation value is $W^R = 0$

Elites in Control, Revolution not Possible

- Suppose state is (μ^l, E)
- No threat of revolution
- Elite will choose not to democratize and zero tax ($\tau = 0$)

- Continuation value to a player of type j is

$$V^j(\mu^l, E) = Ah^j + \delta [(1 - q)V^j(\mu^l, E) + qV^j(\mu^h, E, \tau)]$$

which can be rewritten

$$V^j(\mu^l, E) = \frac{Ah^j + \delta qV^j(\mu^h, E, \tau)}{1 - \delta(1 - q)}$$

Elites in Control, Revolution Possible I

Suppose elites democratize: Continuation value to the poor is

$$V^P(D) = \frac{(1 - \hat{\tau})Ah^p + \hat{\tau}AH}{1 - \delta} = \frac{Bh^p + (A - B)H}{1 - \delta}$$

Suppose elites do not democratize (and $r = 0$), choose tax τ

Continuation value to the poor is

$$\begin{aligned} V^P(\mu^h, E) = & (1 - \tau)Ah^p + \tau AH \\ & + \delta (qV^P(\mu^h, E) + (1 - q)V^p(\mu^l, E)) \end{aligned}$$

Elites in Control, Revolution Possible II

Substituting $V^j(\mu^l, E) = \frac{Ah^j + \delta q V^j(\mu^h, E, \tau)}{1 - \delta(1 - q)}$ into $V^P(\mu^h, E)$ yields

$$V^P(\mu^h, E) = (1 - \tau)Ah^p + \tau AH \\ + \delta \left(qV^P(\mu^h, E) + (1 - q) \frac{Ah^j + \delta q V^j(\mu^h, E, \tau)}{1 - \delta(1 - q)} \right)$$

$$V^P(\mu^h, E)[1 - \delta(1 - q)] = [(1 - \tau)Ah^p + \tau AH][1 - \delta(1 - q)] \\ + \delta(1 - q)Ah^j + \\ + \delta \underbrace{\left[q[1 - \delta(1 - q)] + \delta(1 - q)q \right]}_{=q} V^j(\mu^h, E, \tau)$$

and thus

$$V^P(\mu^h, E) = \frac{(1 - \delta(1 - q))((1 - \tau)Ah^p + \tau AH) + \delta(1 - q)Ah^P}{1 - \delta}$$

Avoiding Revolution Through Redistribution

In state (μ^h, E) poor will not revolt following redistribution (but no democratization) if

$$V^P(\mu^h, E, \tau) \geq W^P \iff$$

$$\frac{(1 - \delta(1 - q))((1 - \tau)Ah^p + \tau AH) + \delta(1 - q)Ah^P}{1 - \delta} \geq \frac{\mu^h AH}{\lambda(1 - \delta)}$$

Payoff to the Poor from redistribution maximized by $\tau = \hat{\tau} = \frac{A-B}{A}$, so democratization will NOT occur iff

$$V^P(\mu^h, E, \tau)|_{\tau=\hat{\tau}} \geq W^P$$

that is,

$$Ah^P + (A - B)H - \delta(1 - q)(A - B)(H - h^P) \geq \frac{\mu^h AH}{\lambda}$$

Revolution and the Commitment Problem

$$Ah^P + (A - B)H - \delta(1 - q)(A - B)(H - h^P) > \frac{\mu^h AH}{\lambda}$$

- One period redistribution only avoids revolution if q (Probability of μ^h) sufficiently large
- Poor have to believe that future redistribution is likely enough
- Rich cannot credibly promise future redistribution without revolutionary threat
- Will only redistribute when $\mu_t = \mu^h$
- If q is too low, Rich will instead democratize when first credibly threatened with revolution

Comparative Statics

$$Ah^P + (A - B)H - \delta(1 - q)(A - B)(H - h^P) \geq \frac{\mu^h AH}{\lambda}$$

- Making B larger makes LHS smaller, so revolution more likely
 - When B gets larger, harder to tax and redistribute
- Making μ^h larger makes RHS larger, so revolution more likely.
 - Costs of revolution is lower

Comparative Statics: Inequality

$$Ah^P + (A - B)H - \delta(1 - q^*)(A - B)(H - h^P) = \frac{\mu^h AH}{\lambda}$$

- Divide both sides by H and rearrange

$$A\frac{h^P}{H} + (A - B) - \delta(1 - q^*)(A - B) + \delta(1 - q^*)(A - B)\frac{h^P}{H} - \frac{\mu^h A}{\lambda} = 0$$

- Implicitly differentiating yields

$$\frac{\partial q^*}{\partial \frac{h^P}{H}} = -\frac{A + \delta(1 - q^*)(A - B)}{\delta(A - B)\left(1 - \frac{h^P}{H}\right)} < 0$$

- Increasing inequality (decreasing $\frac{h^P}{H}$) makes q^* higher, which means revolution is more likely
- Revolution more attractive when there are more resources to be taken